

Retail Business Performance & Profitability Analysis

Project Report | Submitted by: Trisha N

1. Introduction

This project analyzes transactional retail data to understand sales, profitability, inventory behavior, regional performance, and seasonal demand. The project follows the internship brief, which asks analysts to uncover profit-draining categories, optimize inventory turnover, and identify seasonal product behavior.

2. Abstract

The cleaned dataset contains 2,500 unique transactions after duplicate removal. The analysis calculates category and sub-category profit margins, regional and seasonal performance, monthly trends, and the relationship between inventory days and profit. The transaction-level correlation between inventory days and profit is 0.14. Results are provided as CSV analysis tables, SQL queries, Python code, a notebook, and an Excel dashboard.

3. Tools Used

Python (Pandas, NumPy, Matplotlib) for cleaning, EDA and visualization; SQL for aggregation and profitability queries; Excel for a dashboard-ready KPI workbook. The workbook is structured so the analysis can also be recreated in Tableau using the supplied clean CSV data.

4. Steps Involved in Building the Project

- Import the raw retail transaction dataset.
- Remove duplicate Order_ID records and fill missing Discount values with 0.
- Calculate sales, profit, profit margin, average inventory days, and seasonal/monthly metrics.
- Compare categories, sub-categories, regions and seasons to identify performance differences.
- Calculate the correlation between inventory days and profit.
- Create charts and a dashboard-ready workbook for business storytelling.
- Use SQL queries to reproduce the core analysis and identify slow-moving or low-profit areas.

5. Key Findings

- Highest total-profit category in this dataset: Technology (141,129.42). Lowest total-profit category: Office Supplies (41,937.46).
- Highest-sales season: Winter (534,470.36).
- Inventory days vs profit correlation: 0.14; correlation should be treated as an association, not proof of causation.
- Sub-categories with high inventory days and weaker margins are flagged in slow_moving_risk.csv for review.

6. Conclusion

The project converts raw retail transactions into practical business insights. Management can use the results to review slow-moving inventory, monitor discount-related margin pressure, plan seasonal stock, and compare category and regional profitability. The analysis is reproducible through the included SQL file, Python notebook, clean dataset, and Excel dashboard workbook.